DSE 1271 Data Visualization [1 0 3 2]

Lab 10 and 11 Report

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Set- 7

Objective of the analysis:

The main goal of the analysis is to identify trends, patterns, and insights in the movie industry by analyzing various factors, including genre preferences, ratings (including certification), director contributions, IMDb ratings, MetaScores, and other key metrics. This allows movie analysts to observe audience attitudes, critical perceptions, and production trends, etc. over time.

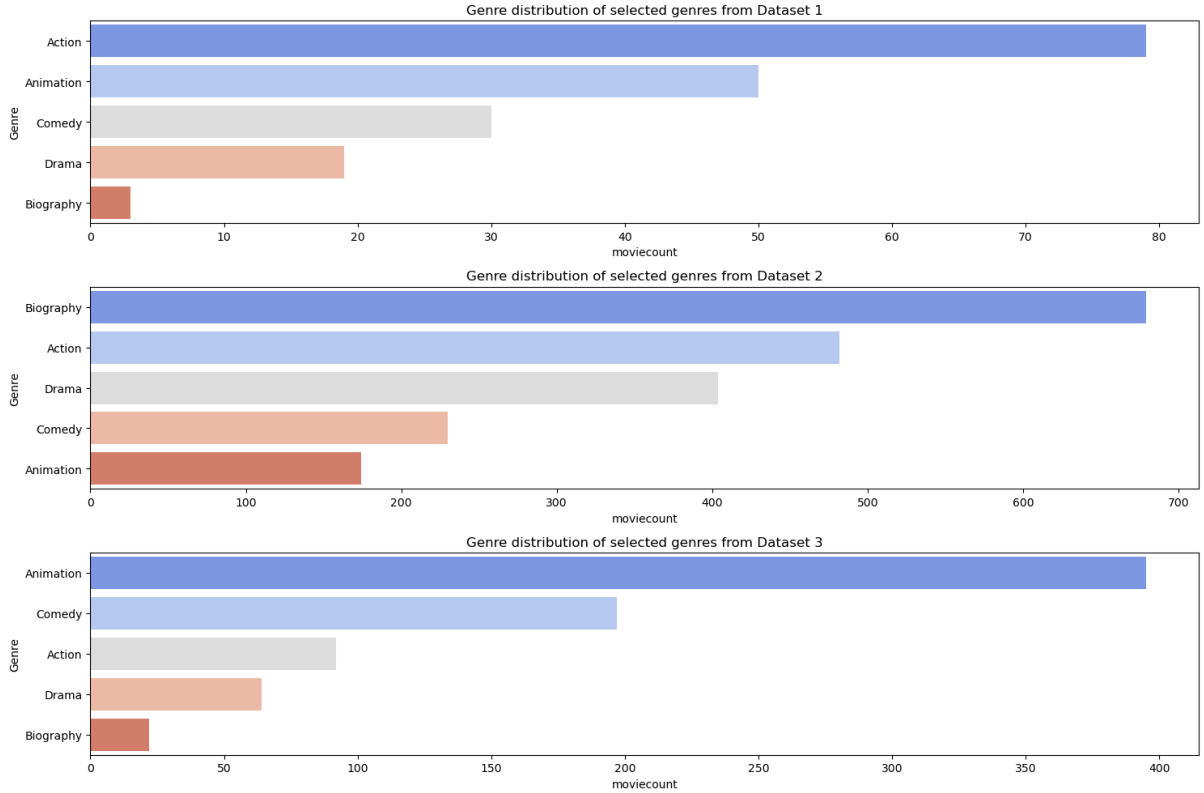
Attributes :

1. **Title**: The name of the movie.
2. **IMDb Rating**: The IMDb rating of the movie.
3. **Year**
4. **Certificate**
5. **Genre** : The genre(s) of the movie.
6. **Director**
7. **Star Cast**
8. **Metascore**: Which is also a rating of the movie summarising critical opinions into a singles score
9. **Duration (minutes)**

The Genres selected are:

1. Action
2. Comedy
3. Animation
4. Biography
5. Drama

Data Exploration:



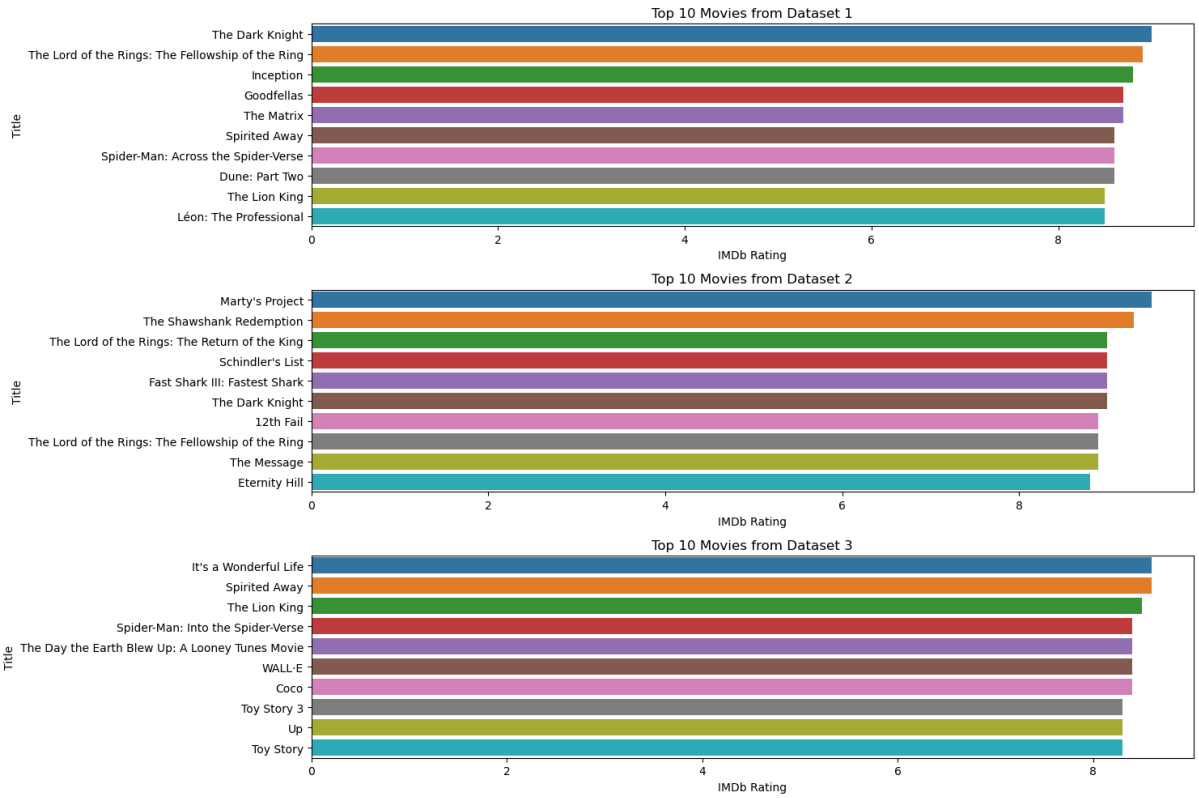
DataSet1: Since it has a large number of action movies followed by animation, the interpretation is that DataSet1 contains all kinds of popular movies.

DataSet2: It includes genres such as biography, documentary, action, adventure, crime, horror, drama, and sci-fi.This dataset reflects a strong interest in real-life stories, possibly driven by an audience seeking inspiration or historical narratives.The relatively lower count of Animation suggests a focus on mature themes rather than family-oriented content. Most of these movies have PG, PG-13, or R ratings, indicating that they are primarily for a mature audience who enjoy **biography or action films.**

DataSet3: While spanning different genres (e.g., documentary, animation, biography, adventure, and comedy) and targeting varied age groups, all the movies in this dataset are rated G, PG, or PG-13, making them suitable for families, including children. Basically **high-energy or family-friendly films.**

- **Dataset 1 leans toward thrill and excitement (Action).**
- **Dataset 2 reflects introspection and realism (Biography).**

- **Dataset 3 emphasizes entertainment and fun (Animation, Comedy).**



Dataset 1: Action and Sci-Fi Dominate

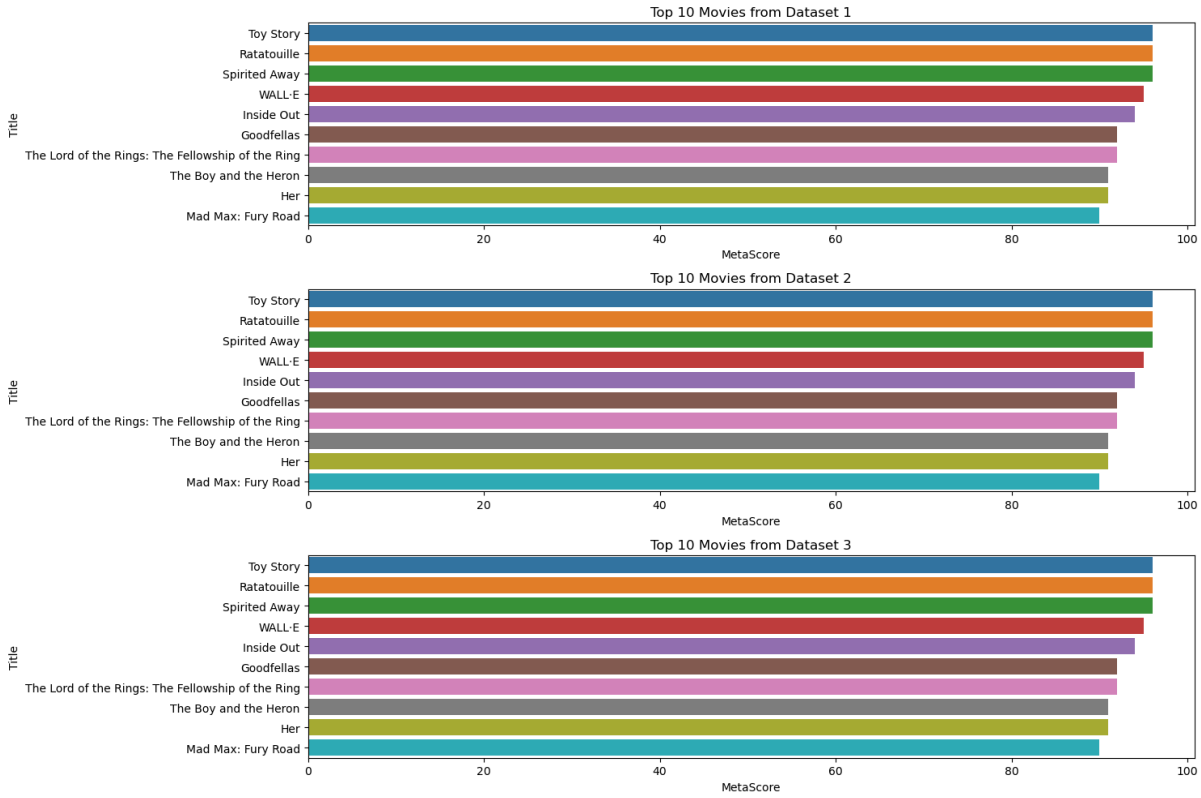
The Dark Knight is followed by The Lord of the Rings: The Fellowship of the Ring and Inception. The list is full of Action, Sci-Fi, and Adventure (The Matrix) and hits for family audiences (The Lion King and Spirited Away), ensuring viewers are captured by compelling storytelling and animation.

Dataset 2: Depth and Drama Take the Spotlight

Marty's Project leads, and then there's the classics like The Shawshank Redemption and The Lord of the Rings: The Return of the King. Biopic dramas features highly in favorites and films with good feelings captivate audiences with a leaning toward serious themes. Audiences give high ratings to story tellers that emotionally connect with them and while they resonate with history and upper crust themes like 12th Fail.

Dataset 3: Animation and Family Films Shine

Animated classics lead, with It's a Wonderful Life, Spirited Away, and The Lion King at the top. Pixar favorites like WALL-E, Coco, and Toy Story dominate, highlighting a strong preference for family-friendly content.Audiences favor uplifting, visually engaging, and heartwarming films, with nostalgia likely influencing high ratings for animated classics.



Dataset 1: Highlights animated films (Toy Story, Ratatouille) and visually stunning adventures (The Lord of the Rings), showing a preference for uplifting, family-friendly, and immersive content.

Dataset 2: Similar trends but with more critically acclaimed dramas (Goodfellas, Her), balancing lighthearted animations with thought-provoking narratives.

Dataset 3: Animation remains dominant (Spirited Away, WALL-E), while action-packed films (Mad Max: Fury Road) indicate an interest in high-energy storytelling alongside family content.

Data Cleaning:

- Integration of all 3 datasets using concat function
- As this DataSet does not contain any NaN values I have opted not to remove NaN.

- There are many duplicate rows in the data frame. To remove them .drop_duplicates function is utilised

	Title	IMDb Rating	Year	Certificates	Genre	Director	Star Cast	MetaScore	Duration (minutes)
0	Gladiator	8.5	2000	R	Action	Ridley Scott	David FranzoniJohn LoganWilliam Nicholson	67.0	155.0
1	Mission: Impossible - Dead Reckoning Part One	7.7	2023	PG-13	Action	Christopher McQuarrie	Tom CruiseHayley AtwellVing Rhames	81.0	163.0
2	Rebel Moon - Part Two: The Scargiver	5.2	2024	PG-13	Action	Zack Snyder	Zack SnyderKurt JohnstadShay Hatten	35.0	122.0
4	Borderlands	7.1	2024	R	Action	Eli Roth	Eli RothJoe Crombie	66.9	116.3
5	Jurassic Park	8.2	1993	PG-13	Action	Steven Spielberg	Michael CrichtonDavid Koepp	68.0	127.0
...	...	...	...	...	...	...	...	...	...
4568	The Boy and the Heron	7.5	2023	PG-13	Animation	Hayao Miyazaki	Soma Santoki	91.0	124.0
4569	Despicable Me 4	6.3	2024	PG	Animation	Chris Renaud	Steve Carell	50.0	95.0
4570	Inside Out	8.1	2015	PG	Animation	Pete Docter	Amy Poehler	94.0	95.0
4571	IF	6.5	2024	PG	Animation	John Krasinski	Cailey Fleming	46.0	104.0
4572	Inside Out 2	7.9	2024	PG	Animation	Kelsey Mann	Amy Poehler	73.0	96.0

2830 rows × 9 columns

- The **Star Cast** in the dataframe is so clumsy that we cannot perform anything on it so I split the Star Cast in respect to the Capital letter and does not follow a space.

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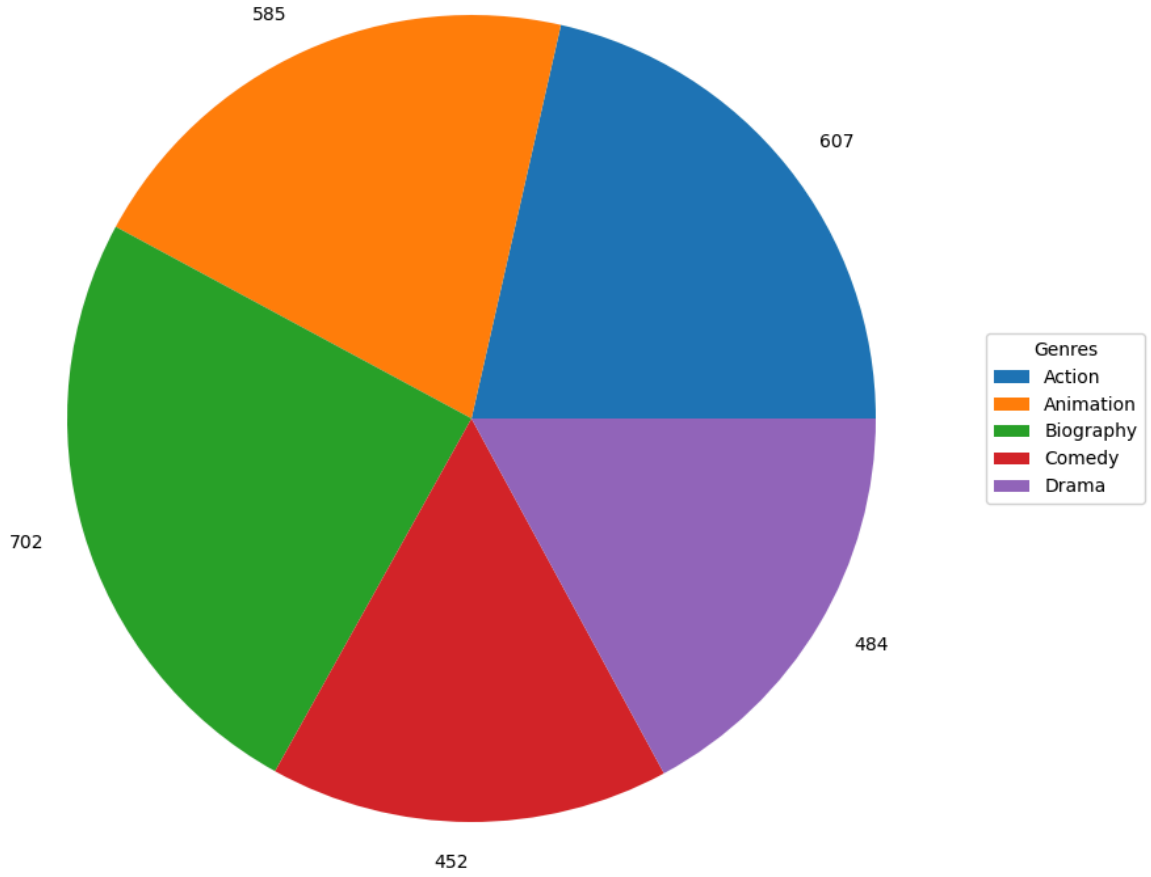
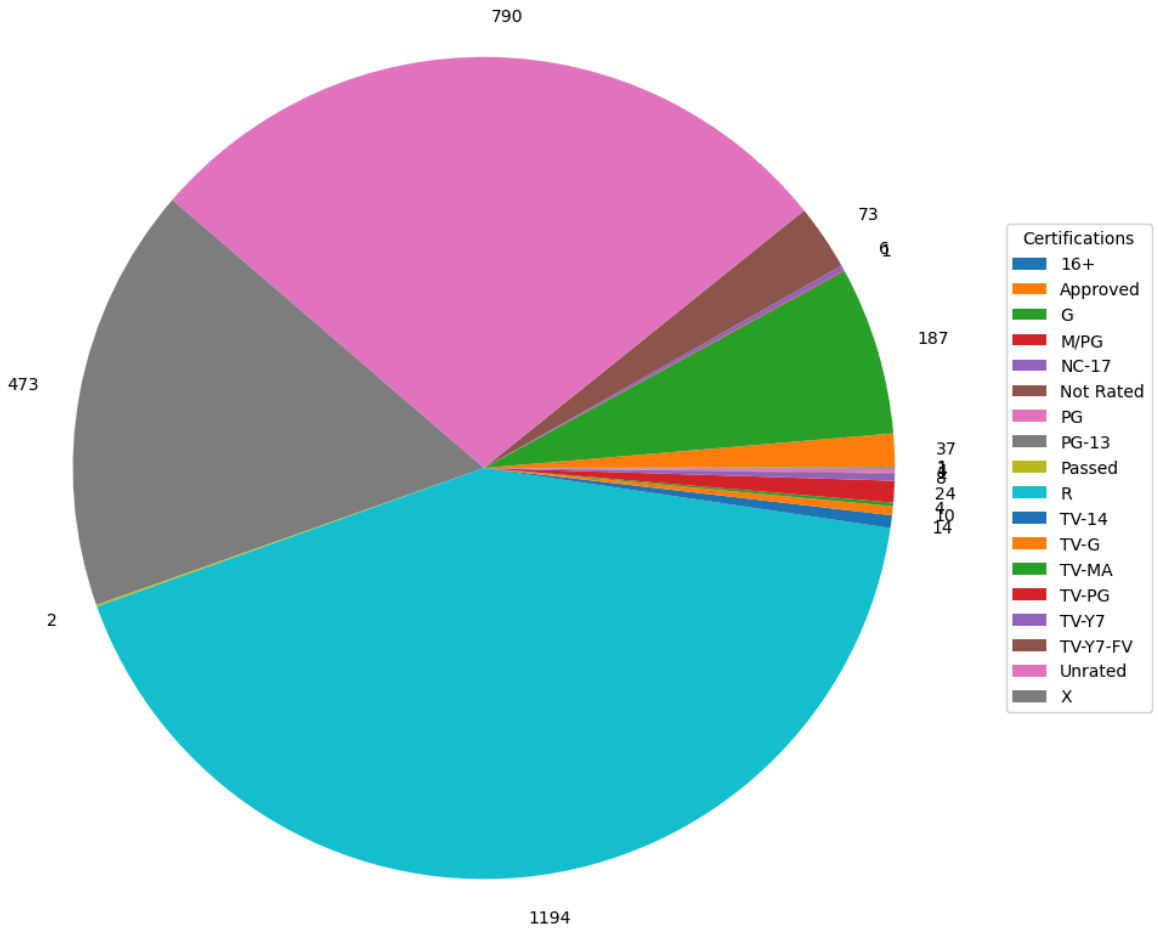
6782 rows × 9 columns

Analysis and Insights:



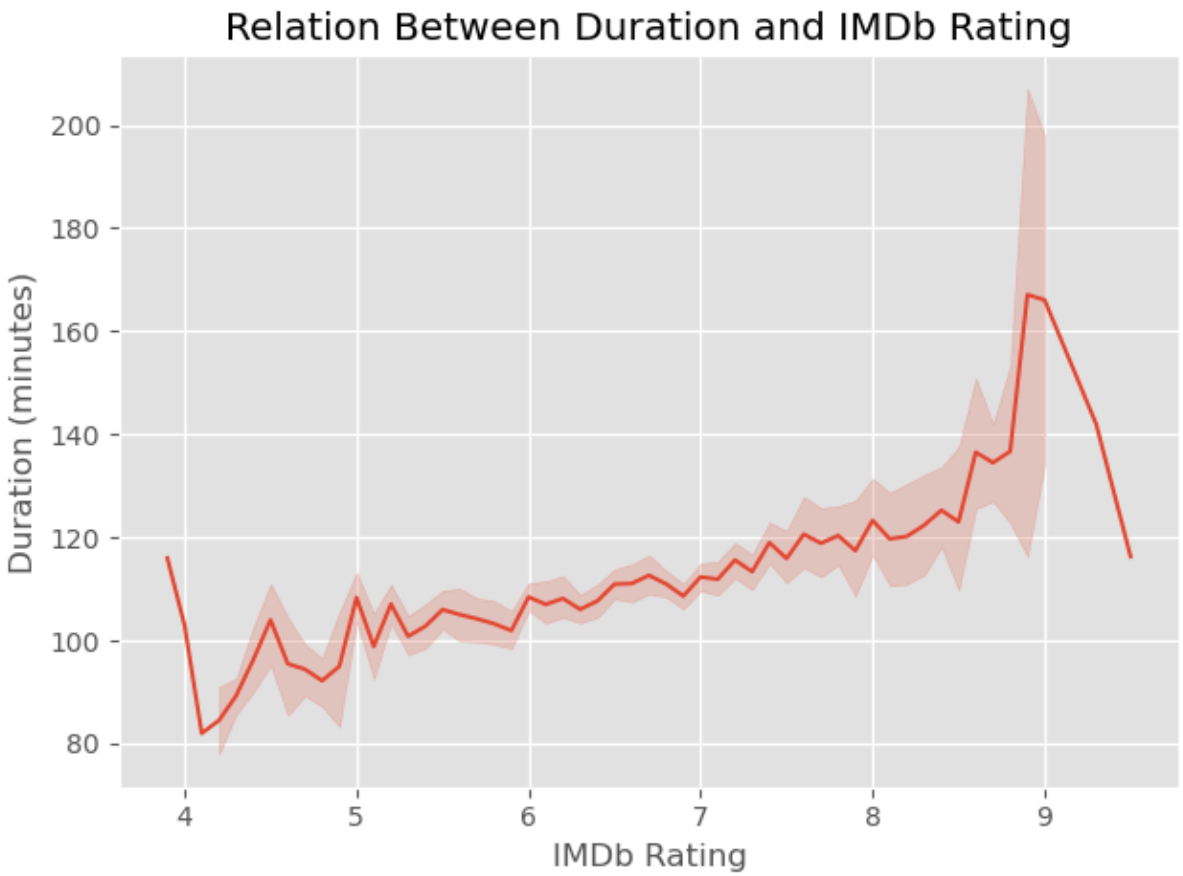
The pie chart illustrates the distribution of movies across five genres: Action, Animation, Biography, Comedy, and Drama.

1. Biography (702 movies): Most represented genre.
    - Reflects high interest in real-life stories.
2. Action (607 movies):Second largest share.
    - Highlights universal appeal for thrilling content.
3. Animation (585 movies): Strong representation.
    - Popular for family-friendly and visually engaging films.
4. Drama (484 movies): Moderate presence.
    - Appeals to audiences seeking emotional depth.
5. Comedy (452 movies): Least represented.
    - Still significant for lighthearted entertainment.

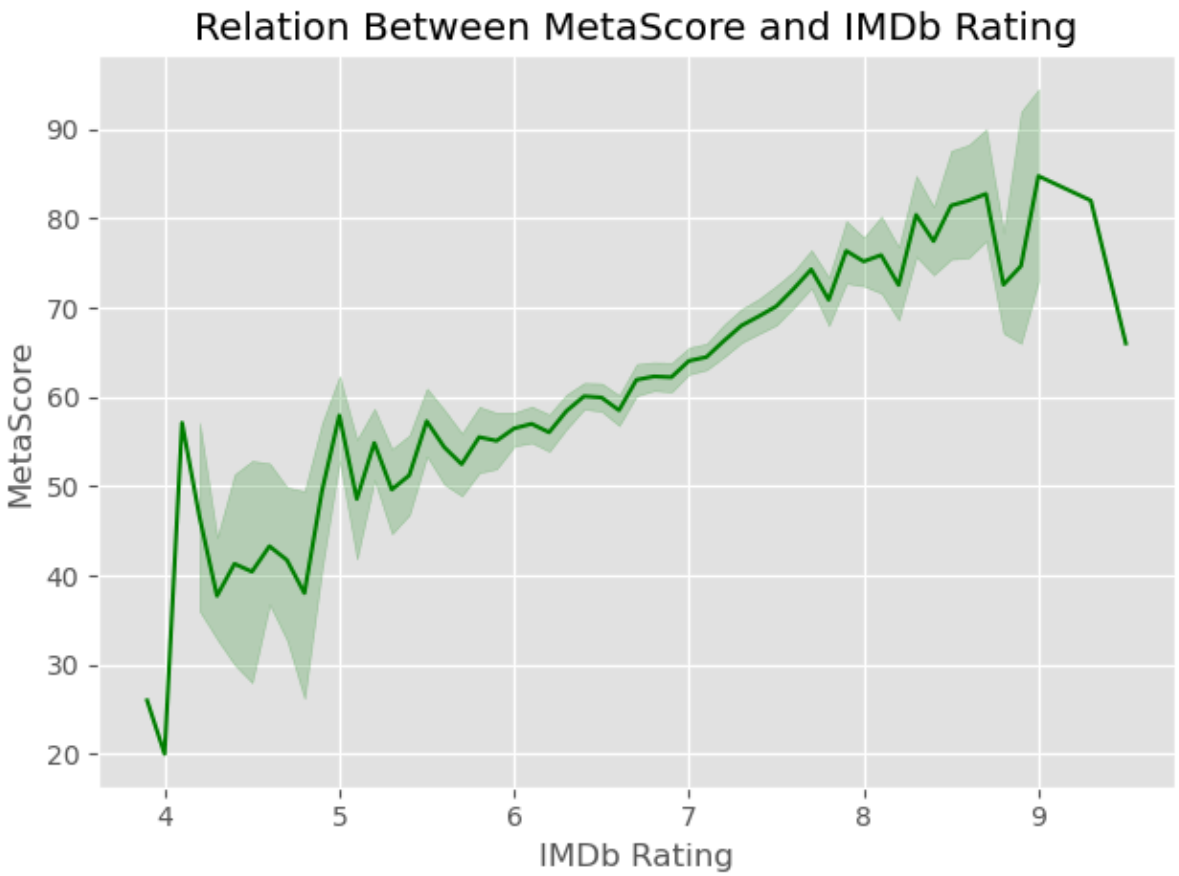


The pie chart illustrates the distribution of movie certifications, providing insights into the type of content being produced and its target audience.

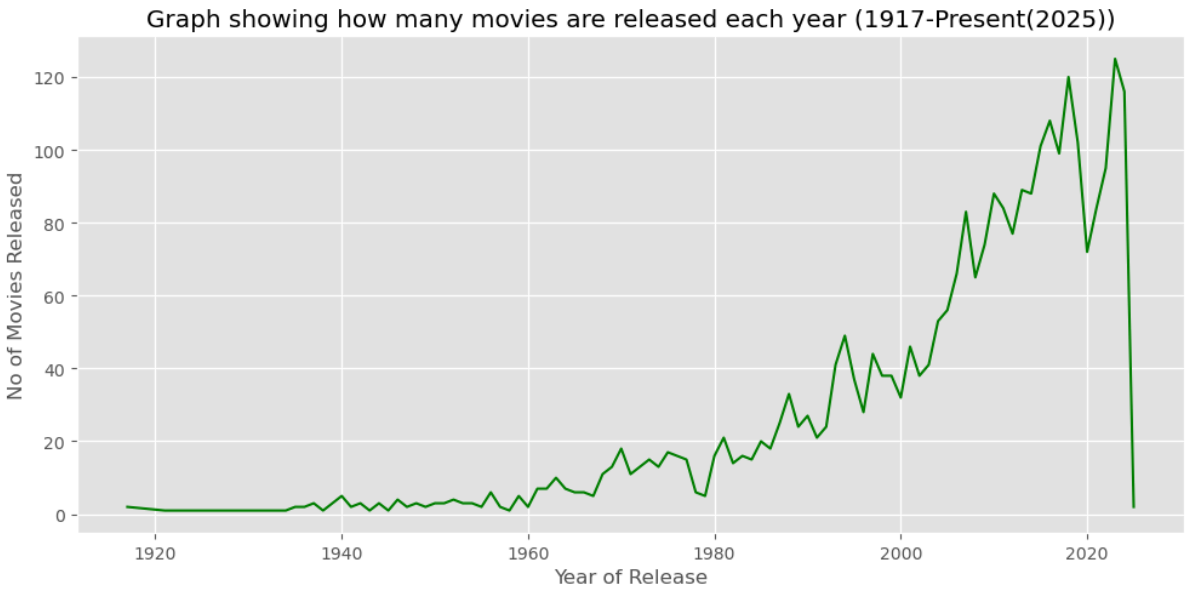
- R (1,194 movies): Highlights a significant proportion of movies targeting mature audiences due to strong language, violence, or adult themes.
- PG (790 movies):The second largest category, emphasizing family-friendly content.
- PG-13(473 movies): indicating a strong focus on content suitable for teens and families with some restrictions.
- Smaller shares include:
    - G (187 movies)
    - Not Rated (73 movies)
    - TV-G (37 movies).

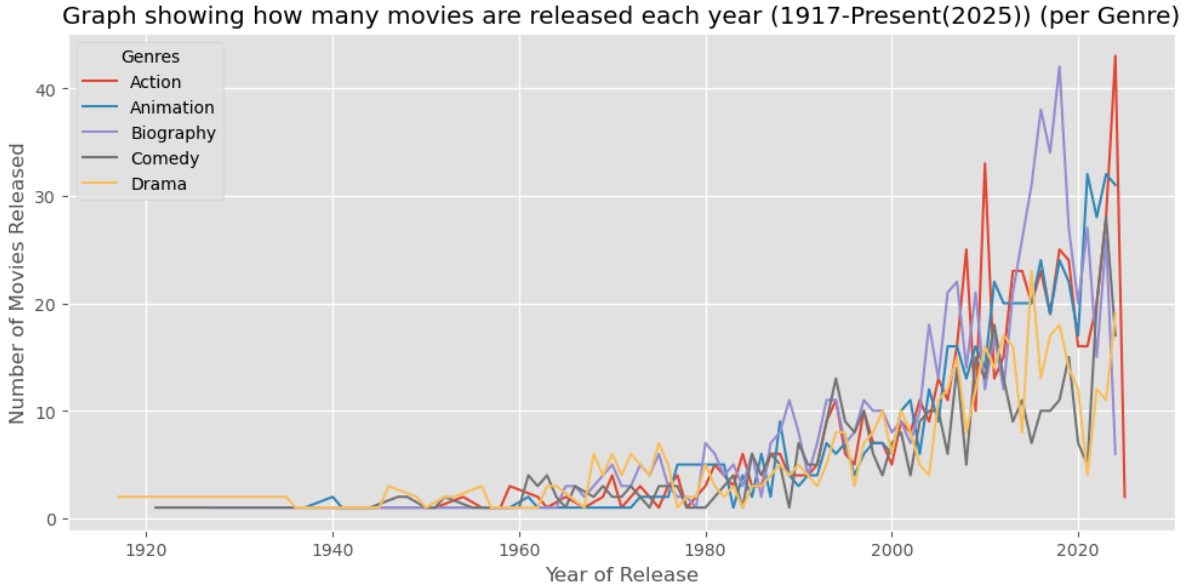


- As the IMDb rating increases, the average movie duration also tends to increase.
- Movies with higher ratings (8–9) generally have longer durations, often exceeding 140 minutes.
- A few highly rated movies (above 9) show a sharp drop in duration, indicating shorter films can also achieve high ratings.
- Movies with lower ratings (4–5) tend to have shorter durations, typically around 90 minutes or less
- Longer movies are often associated with higher IMDb ratings, suggesting that audiences may value detailed storytelling and character development.
- Shorter movies dominate lower ratings, potentially indicating less engagement or rushed narratives.



- A clear positive trend is observed: as IMDb ratings increase, MetaScores also tend to rise.
- This suggests that movies rated highly by audiences are generally well-received by critics.
- A slight drop in MetaScores is observed for IMDb ratings above 9, suggesting that while audiences may rate these movies exceptionally high, critics may not always align

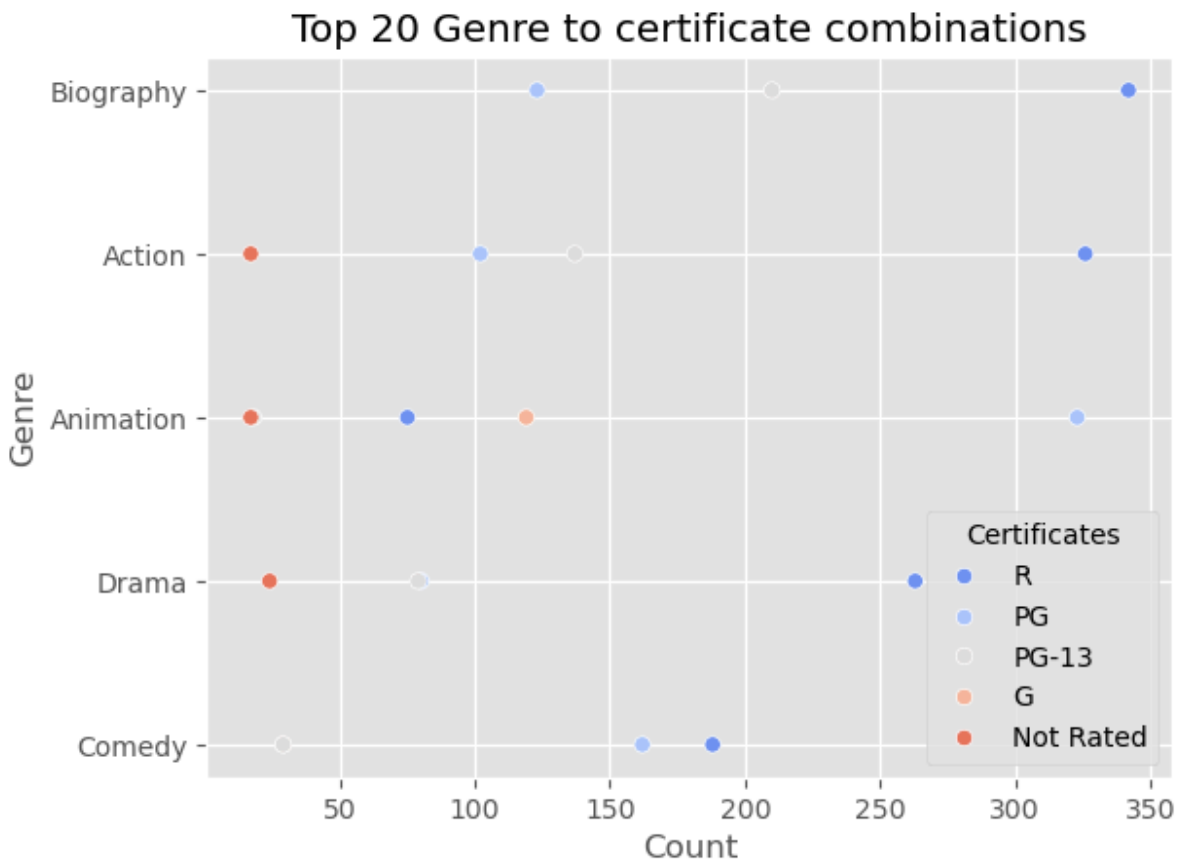




- Movie releases increased steadily from the 1920s to the early 2000s, reflecting the growth of the film industry.
- A significant increase in movie releases is observed after 2000, peaking around 2020. This could be attributed to advancements in technology and the rise of streaming platforms.
- A sharp decline is visible in 2025, likely due to incomplete data

From per Genre graph

- Action and Drama Dominate:Action and Drama genres have consistently seen higher releases compared to others, especially post-2000.
- Animation's Rise: Animation shows a noticeable increase in releases after the 1990s, likely driven by advancements in CGI and family-friendly content demand.
- Biography's Growth: Biography movies have seen a surge in recent years, reflecting an audience interest in real-life stories.



Biography: Has associated with R certification, with over 325 movies.Few Biography movies are rated PG or PG-13, indicating a focus on broader audience appeal.

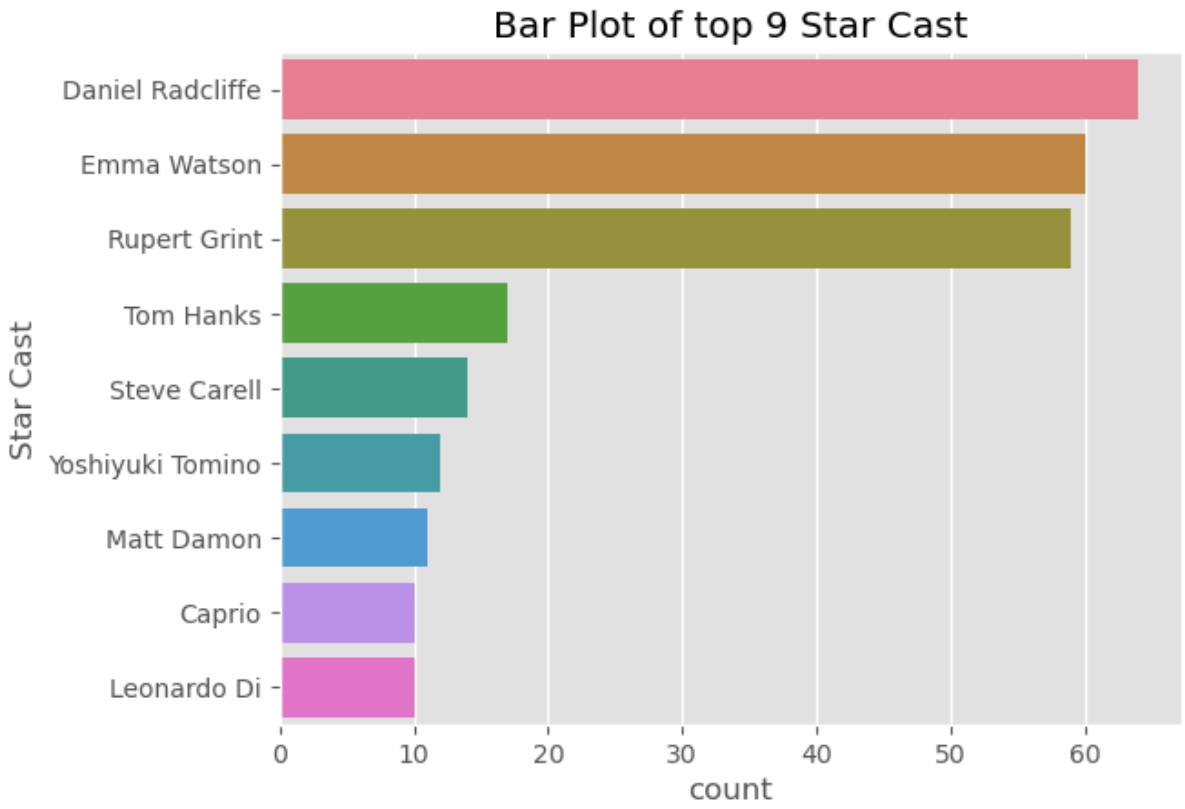
Action: Balanced distribution across PG-13 and PG-13 certifications, with R leading at around 315 movies.Minimal representation of Not-rated Action films, reflecting the genre's mature themes.

Animation: Strong presence in PG and G certifications, with counts exceeding 300 and 100 respectively for both.Rarely associated with R certification, emphasizing its family-friendly nature.

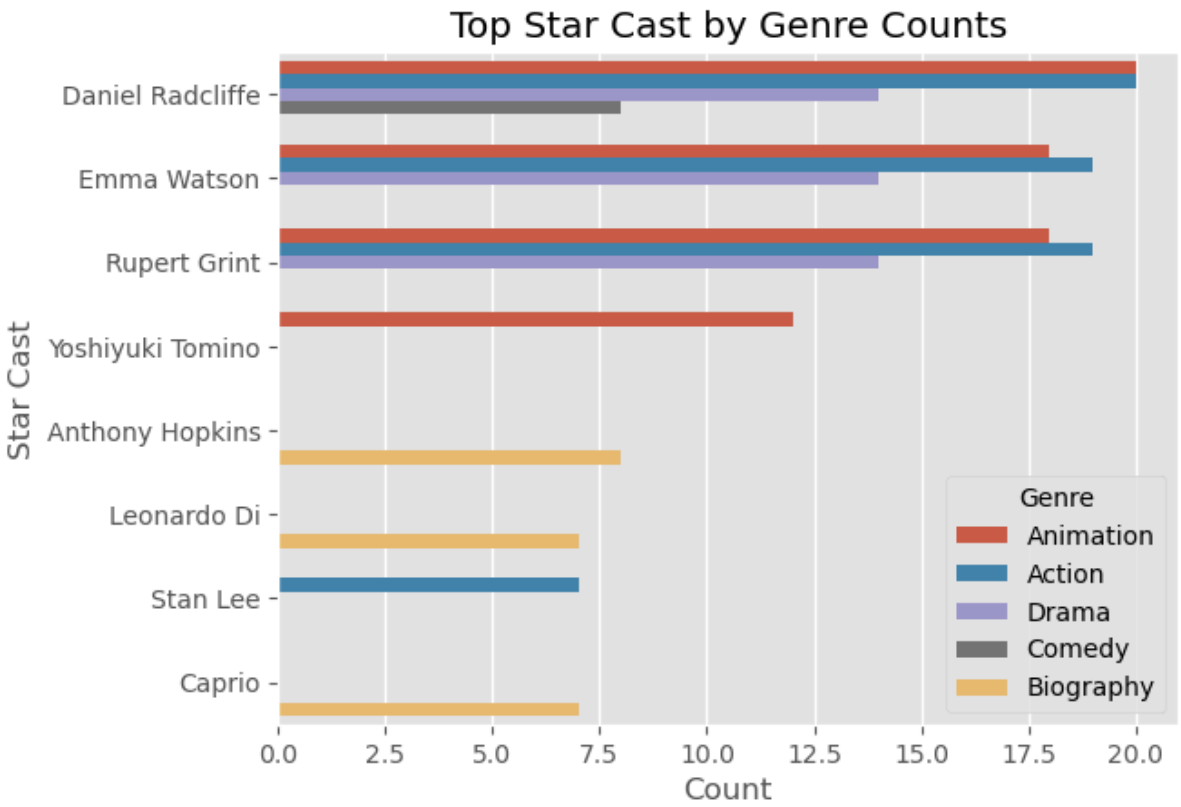
Drama: Significant representation in R certification (>250 movies), reflecting mature and emotionally intense content. Moderate presence in PG-13 and Not Rated categories.

Comedy: Mostly distributed across R and PG certifications, with counts around 100–150. Few Comedy movies are rated R or Not Rated, aligning with its lighthearted appeal.

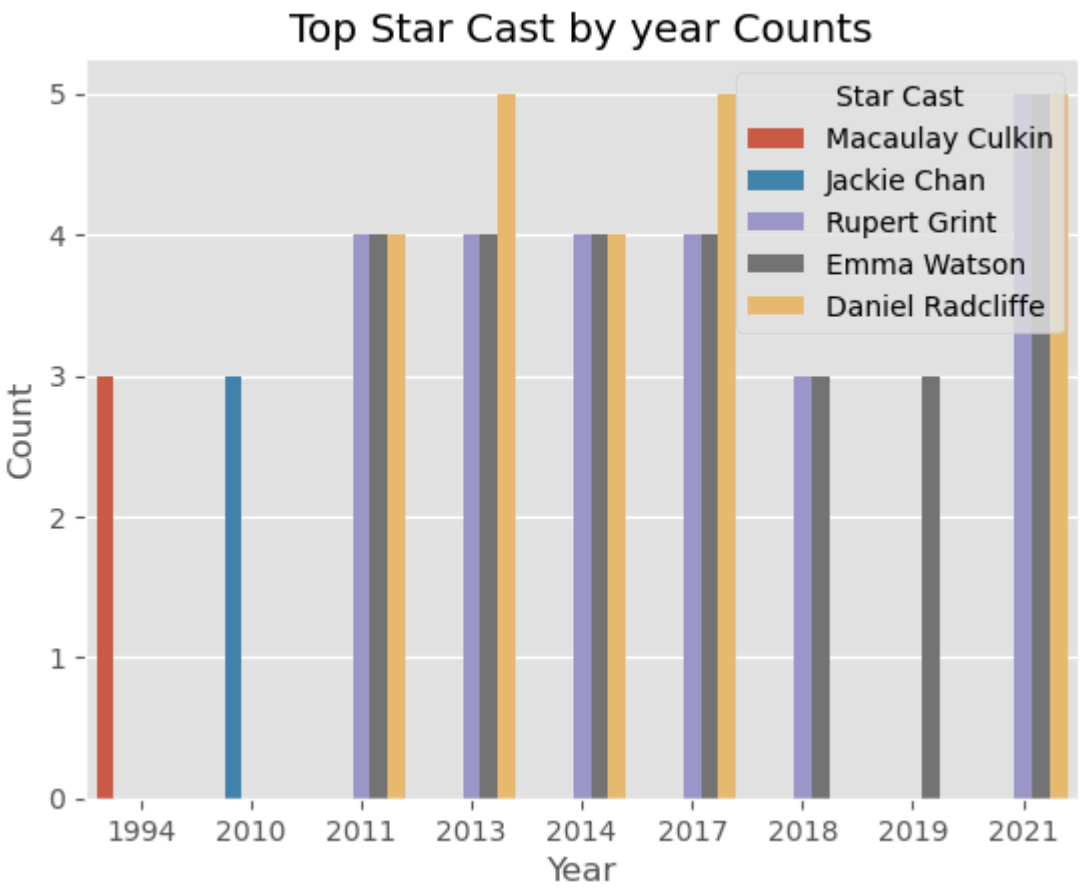
Star Cast:



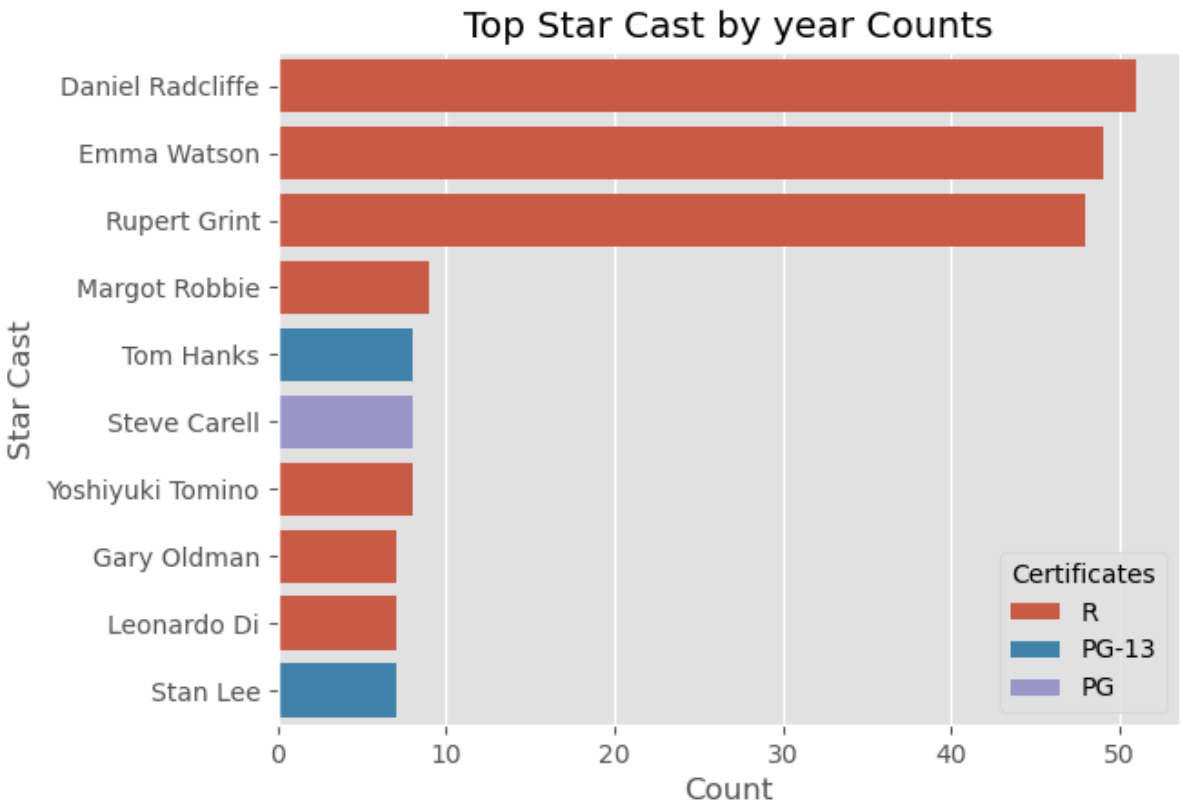
1. Daniel Radcliffe Leads: With over 60 appearances, Daniel Radcliffe dominates the list, likely due to his iconic role in the Harry Potter franchise.
2. Emma Watson and Rupert Grint Follow: Both actors, also from Harry Potter, rank second and third, showcasing the franchise's significant contribution to their filmography.
3. Other Stars: Tom Hanks, Steve Carell, and Leonardo DiCaprio feature prominently, reflecting their versatility and popularity across diverse genres.
4. The dominance of Harry Potter stars highlights the lasting impact of blockbuster franchises on actors' careers.



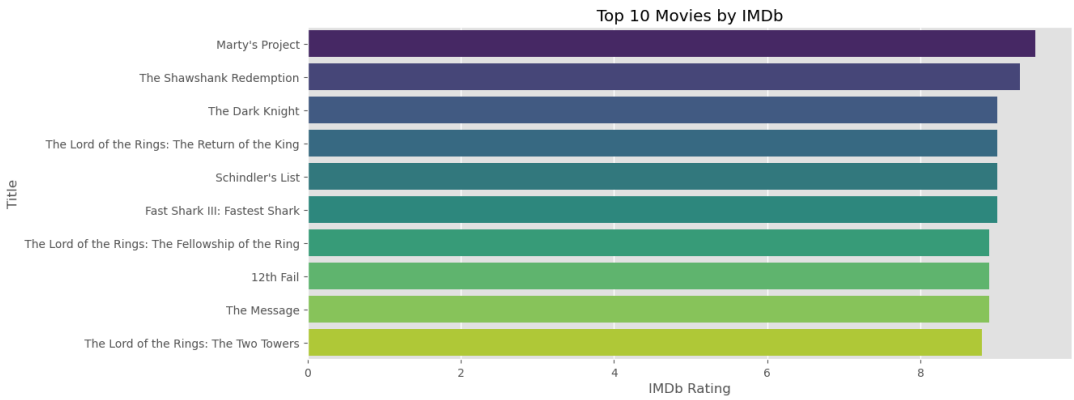
1. Daniel Radcliffe Leads: Dominates across genres, especially in Drama and Action, reflecting his versatility beyond the Harry Potter franchise.
2. Emma Watson and Rupert Grint: Prominent in Drama and Comedy, showcasing their strong association with the Harry Potter series and other projects.
3. Yoshiyuki Tomino: Primarily featured in Animation, highlighting his niche expertise in animated films.
4. The chart emphasizes the dominance of actors associated with major franchises and their genre-specific contributions.

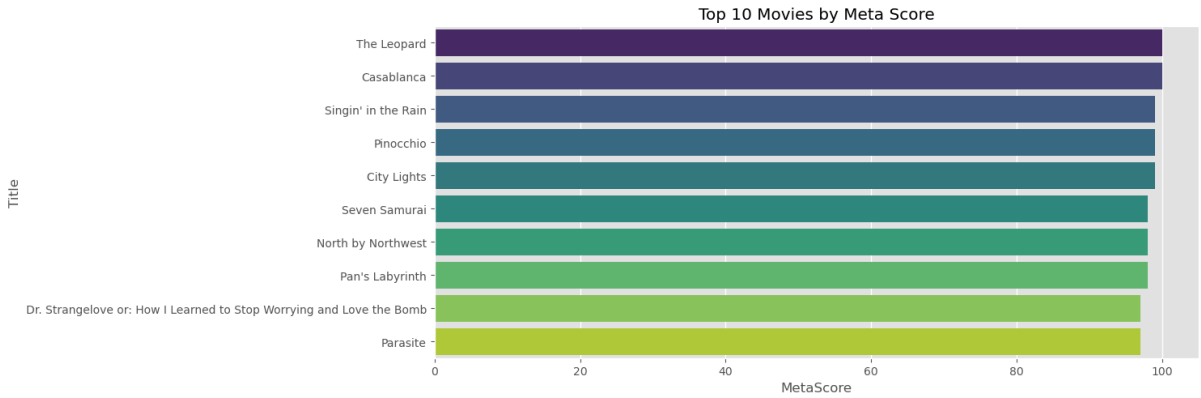


1. Emma Watson,Rupert Grint and Daniel Radcliffe Lead: These actors dominate the chart with consistent appearances across multiple years, reflecting the enduring popularity of the Harry Potter franchise.
2. Macaulay Culkin's Peak: His contributions are primarily concentrated in earlier years, likely due to his iconic roles in Home Alone films.
3. Jackie Chan's Steady Presence: Jackie Chan shows consistent appearances, highlighting his global appeal across action and comedy genres.
4. Key Insight: Franchise stars like Emma Watson ,Rupert Grint and Daniel Radcliffe maintain a strong presence over time, while actors like Macaulay Culkin peak during specific periods.



1. Daniel Radcliffe Leads: Daniel Radcliffe has the highest count, reflecting his consistent appearances, likely due to the Harry Potter franchise's success.
2. Emma Watson and Rupert Grint Follow: Both actors rank second and third, showcasing their significant contributions to the same franchise.
3. Diverse Certifications: The chart highlights a mix of PG, PG-13, and R-rated movies across the top actors, indicating their versatility in catering to different audiences.





IMDb vs. MetaScore Analysis

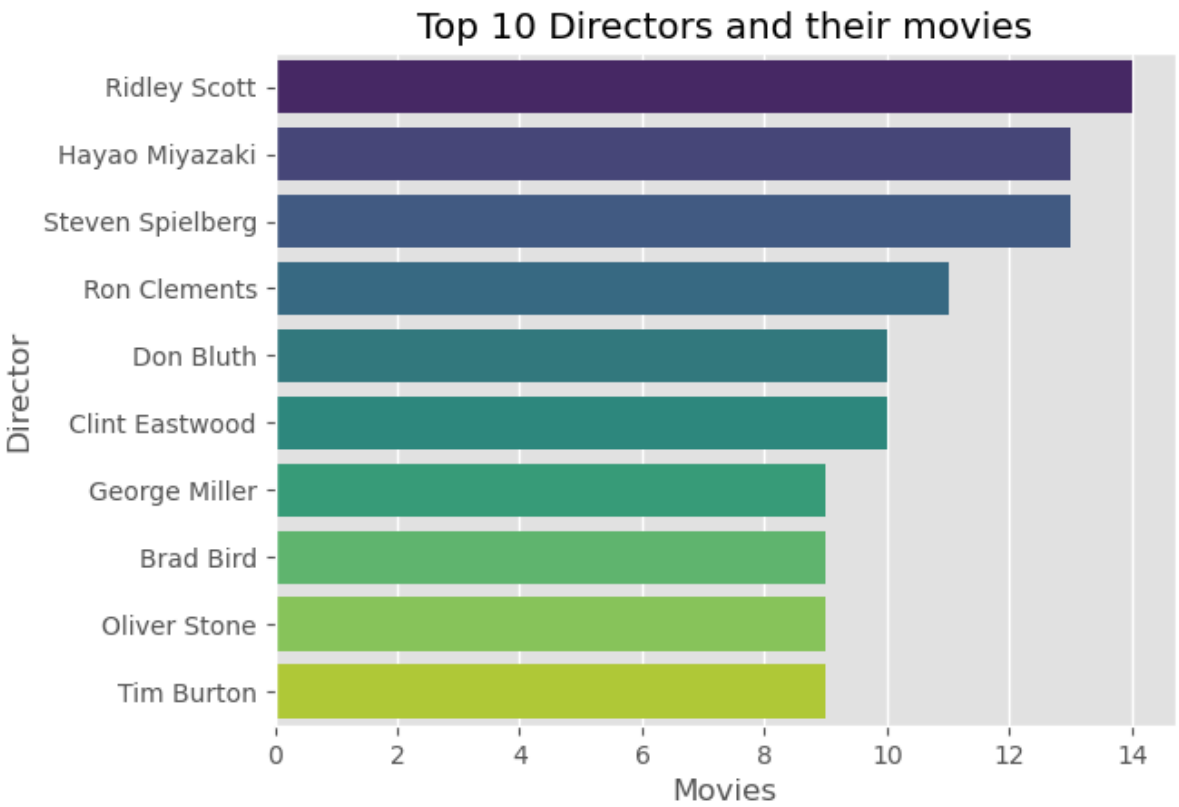
- Audience Favorites: High-rated films like The Shawshank Redemption and The Dark Knight (IMDb ~9) show broad appeal.
- Genre Diversity: Top IMDb films span drama (Schindler's List), fantasy (LOTR), and action (Fast Shark III)
- Franchise Success: The Lord of the Rings films highlight blockbuster dominance.
- Critics' Choice: Classics like The leopard, Casablanca and Singin' in the Rain (MetaScore ~100) reflect critical acclaim
- Artistic Preference: MetaScores favor timeless classics and international films (Parasite, Pan's Labyrinth)
- IMDb favors modern blockbusters, while MetaScores value artistry and legacy. Overlaps (e.g., LOTR) indicate universal acclaim.

	Director	Star Cast	Count
5851	Steven Spielberg	Tom Hanks	4
1973	Genndy Tartakovsky	Adam Sandler	3
4690	Peter Jackson	J.R.R. Tolkien	3
635	Bill Corbett	Rupert Grint	3
634	Bill Corbett	Emma Watson	3
633	Bill Corbett	Daniel Radcliffe	3
2642	Jeff Fowler	Ben Schwartz	3
282	Andrew Erwin	Jon Erwin	3
1461	Dean DeBlois	Jay Baruchel	3
6479	Zack Snyder	Shay Hatten	3

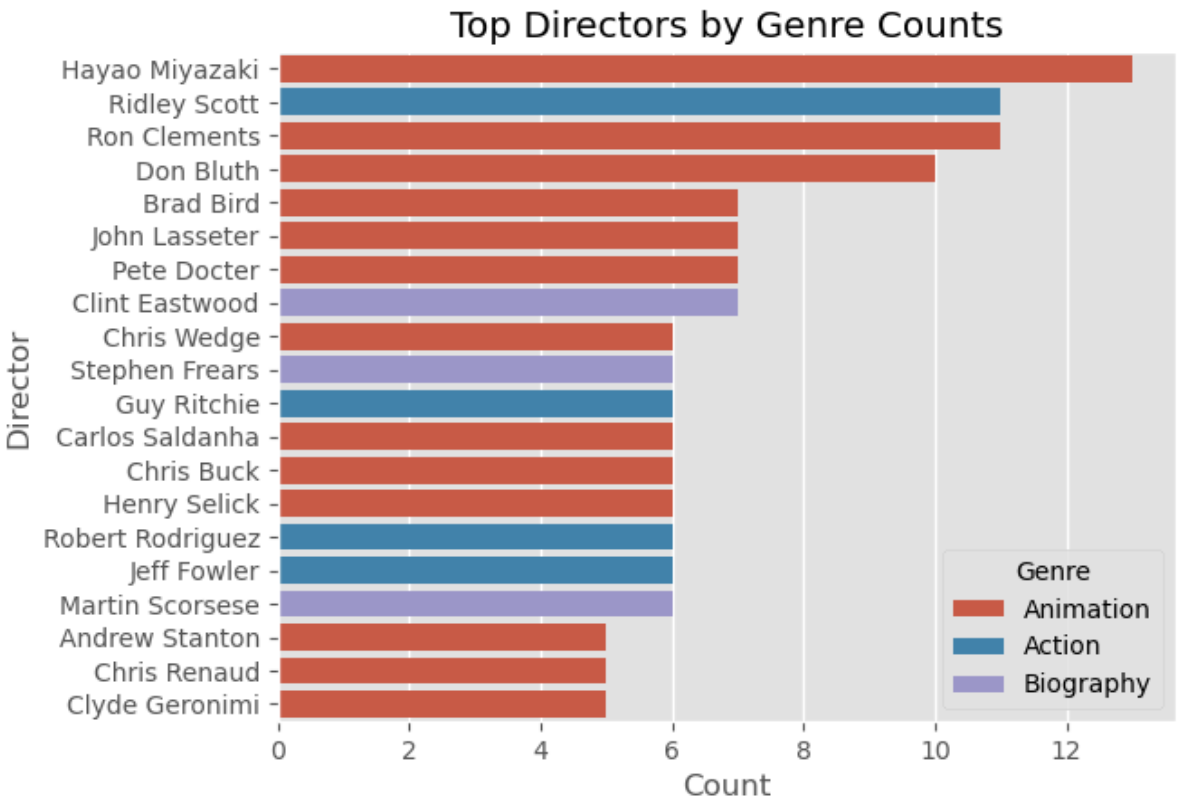
Top 10 Director-Actor Collaborations

- Spielberg & Hanks (4 films): A powerhouse duo behind Saving Private Ryan and Catch Me If You Can.
- Tartakovsky & Sandler (3 films): Known for animated hits like Hotel Transylvania.
- Jackson & Tolkien (3 films): The Lord of the Rings trilogy cements their legacy.
- Corbett & Harry Potter Trio (3 films each): Worked with Grint, Watson, and Radcliffe, showcasing franchise continuity.
- Other Notable Pairs:
    - Jeff Fowler & Ben Schwartz (animation)
    - Andrew & Jon Erwin (drama)
    - Zack Snyder & Shay Hatten (action)
- Strong director-actor collaborations fuel successful franchises and critically acclaimed films, proving the power of creative synergy.

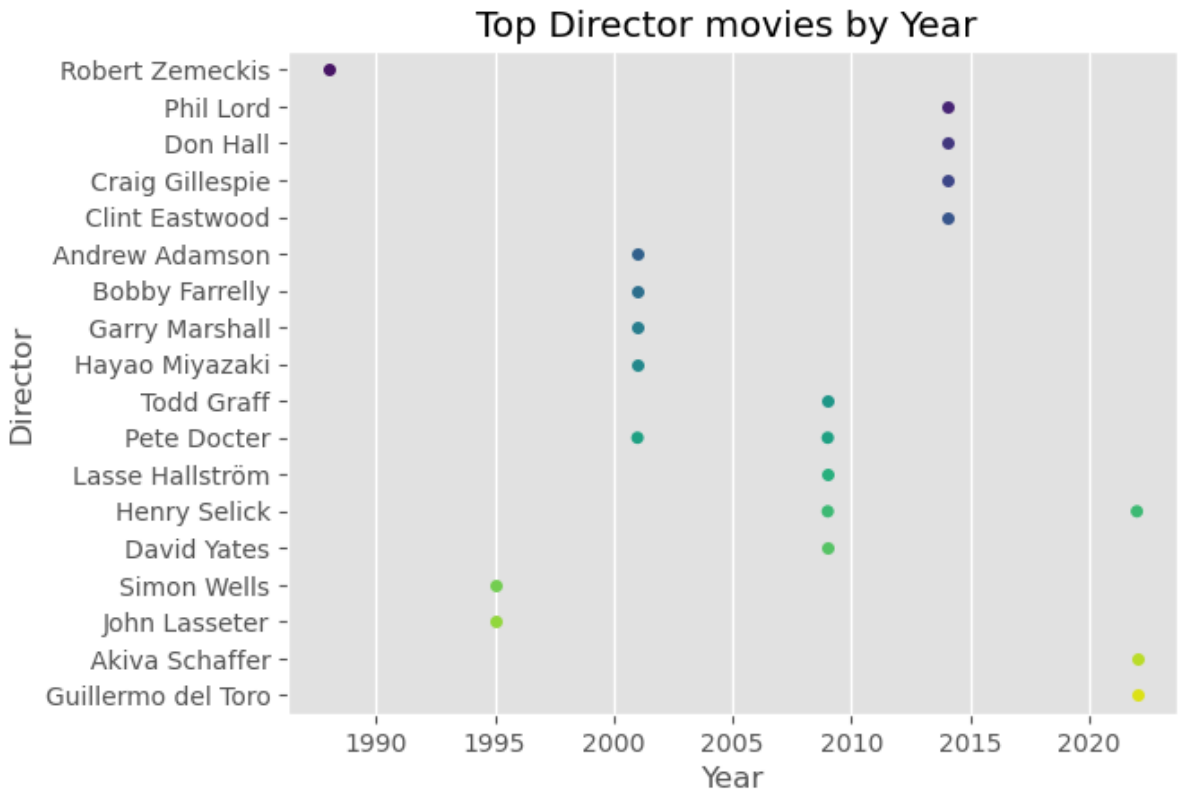
Director Statistics



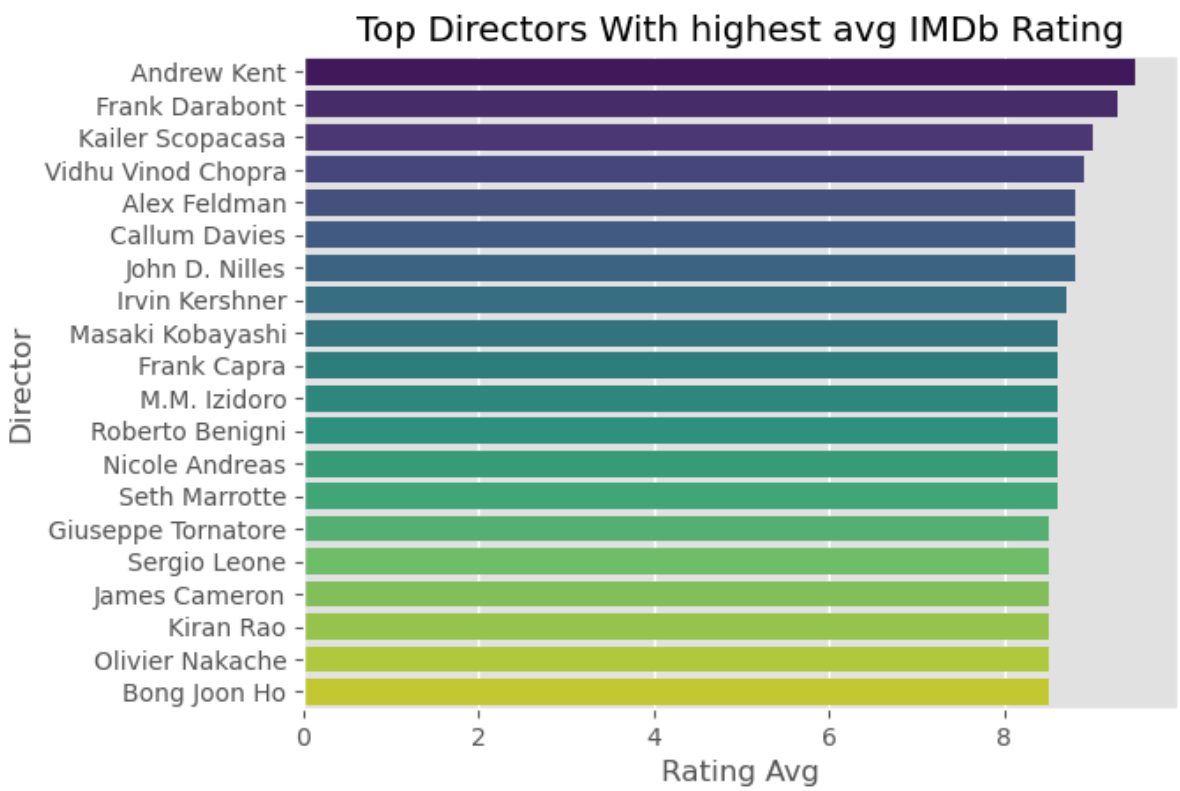
1. Ridley Scott Leads: With 14 movies, Ridley Scott dominates the list, showcasing his prolific career in directing iconic films across genres like Gladiator and Alien.
2. Hayao Miyazaki Follows: Known for his animated masterpieces (Spirited Away, My Neighbor Totoro), Miyazaki secures the second spot with 12 movies.
3. Steven Spielberg: Ranking third with 11 movies, Spielberg's diverse portfolio includes blockbusters like Jurassic Park and E.T.
4. Ron Clements and Don Bluth: Both directors are tied with 10 movies each, highlighting their contributions to animation and family-friendly films.
5. Other Notables: Clint Eastwood, George Miller, Brad Bird, Oliver Stone, and Tim Burton round out the list with 8–9 movies each, reflecting their consistent impact on the industry.
6. The chart highlights the influence of directors with diverse portfolios across animation, drama, action, and adventure genres.



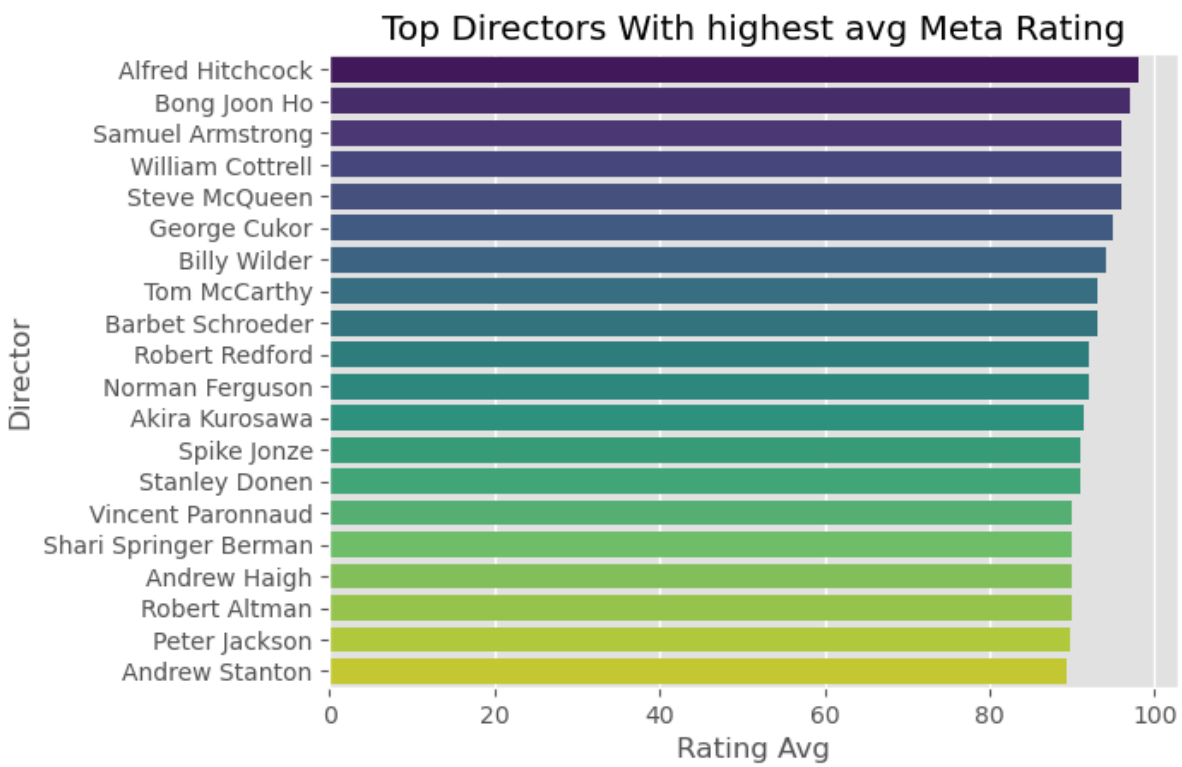
1. Hayao Miyazaki Leads: Dominates with Animation, showing his expertise in creating iconic animated films like Spirited Away.
2. Ridley Scott Follows: Known for Action films, he ranks second, showcasing his impact on the genre (Gladiator, Alien).
3. Animation Dominance: Directors like Ron Clements, Don Bluth, and John Lasseter highlight Animation's strong representation.
4. Action and Biography: Clint Eastwood and Martin Scorsese contribute significantly to Action and Biography genres, reflecting their versatility.
5. The chart emphasizes the influence of directors across Animation, Action, and Biography genres in shaping the industry.



1. This Graph represents that each director has made 2 films in the adjacent year
2. Consistent Contributions: Directors like Hayao Miyazaki, Clint Eastwood, and Guillermo del Toro show consistent movie releases over multiple decades, reflecting their sustained impact on the industry.
3. Recent Activity: Directors such as Phil Lord and Don Hall are more active in recent years, highlighting their relevance in current filmmaking trends.
4. Genre Diversity: The chart includes directors from various genres, showcasing a broad spectrum of creative storytelling across years.
5. The timeline demonstrates the evolving contributions of both classic and modern directors, emphasizing their influence on different eras of cinema.



1. Andrew Kent Leads: Tops the chart with the highest average IMDb rating, reflecting consistent audience acclaim for his films.
2. Frank Darabont and Kailer Scopacasa Follow: Both directors rank highly, showcasing their ability to deliver critically acclaimed movies.
3. International Representation: Directors like Bong Joon Ho and Olivier Nakache highlight the global appeal of high-quality storytelling.
4. Genre Diversity: The list includes directors excelling in various genres, from drama to animation, emphasizing their versatility.
5. Consistently high IMDb ratings indicate these directors' ability to resonate with both critics and audiences alike.



1. Alfred Hitchcock Leads: Tops the list, showcasing his consistent critical acclaim for suspenseful and iconic films.
2. Bong Joon Ho and Samuel Armstrong: Follow closely, reflecting their ability to deliver highly rated films across genres.
3. Animation Excellence: Directors like Andrew Stanton and Peter Jackson highlight the strong presence of animated and fantasy films in the top rankings.
4. Genre Diversity: The list includes directors excelling in suspense, drama, animation, and fantasy, emphasizing their versatility.
5. High MetaScores indicate these directors' ability to resonate strongly with critics through artistic and impactful storytelling.

Conclusion:

- In the genres I have chosen(Action, comedy , Animation , Biography and Drama), R, PG-13 and PG certifications dominate, indicating a focus on Mature audience, family-friendly content.
- Franchise directors and cast such as those from Harry Potter show significant impact on industry trends.
- MetaScores emphasize critical acclaim for classics and international films.
- High IMDb ratings correlate with longer movie durations
- Movie releases have surged post-2000 due to technological advancements and streaming platforms.
- Animation and Biography genres have shown significant growth in recent years.
- The analysis highlights the importance of balancing audience appeal with critical quality to achieve long-term success in the movie industry.